Team 112 Mid Submission

Student

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1 Introduction

2 Current Approach

2.1 Product Overview

Gullak is an offline, single-player Android game designed to teach financial literacy to Grade 11–12 students in Hyderabad, Telangana. The application uses a connected narrative titled *The Money Diaries*, available in English and Telugu. Players follow Asha, a Class 11 student who is saving money towards her share of a laptop for a Youth Innovation Fair.

All money used in the game is fictional. The application does not require login credentials, network connectivity, or personal data.

2.2 Learning Design

Each of the seven chapters follows a consistent learning structure: an illustrated dialogue, an interactive activity, a decision with visible consequences, a short learning card, and a recap.

Chapter	**Topic**	**Core Interaction**
1. The First Rs. 1,000	Budgeting, needs vs. wants	Add and remove expenses against Rs. 1,000 while tracking the live balance.
2. The Laptop Dream	Savings goals and regular saving	Choose a target and decide weekly savings over four weeks.
3. Welcome to Digital Money	Bank balances, UPI, and pending payments	Use a simulated bank to deposit, withdraw, make payments, and check the passbook.
4. The Scholarship Message	Phishing and OTP/PIN safety	Inspect four messages, identify warning signs, and decide how to respond.
5. Buy Now or Wait?	Debit vs. credit and interest	Compare three options, complete a checkout, and explore a credit statement.

Chapter	Topic	Core Interaction
6. Make Your Money Work	Interest and liquidity	Compare 3% savings interest with a 6% fixed deposit over six months.
7. When Plans Change	Emergency funds and insurance	Read a policy and rebuild a household financial plan after a Rs. 1,000 bill.

2.3 Connected Simulation

Financial decisions and balances carry forward between chapters instead of resetting after every activity.

- Leftover pocket money from Chapter 1 becomes the savings jar and subsequently the bank balance in Chapter 3.
- The saving habit established in Chapter 2 determines the monthly savings amount used in Chapter 6.
- Scam losses in Chapter 4 and store credit taken in Chapter 5 affect subsequent chapters and the ending.
- Poor financial decisions have consequences, but the game always provides a recovery path and does not block story completion.

2.4 Technical Approach

The application uses Flutter and Dart for development, Riverpod for state management, SQLite through `sqflite` for progress persistence, and bundled JSON files for story text. No backend is used.

- **Single source of truth:** An immutable `StoryState` per profile stores balances, decisions, and checkpoints. Screens report player actions, while a pure-Dart rules module recomputes and validates changes.
- **Data-driven chapters:** Each chapter consists of beats representing dialogue, activities, decisions, lessons, and recaps. Conditional beats allow previous choices to alter later scenes.
- **Internationalisation:** Approximately 540 story strings per language are stored in JSON, allowing content and translations to be changed independently of the application code.
- **Persistence:** A checkpoint is saved after every beat. A database revision counter prevents duplicate saves, while the controller ignores taps on outdated beats.
- **Replay:** Completed chapters can be replayed for practice from their initial state without modifying the saved story.
- **Scoring:** Stars reward sound financial decisions, such as planning correctly and verifying transactions, rather than the amount of money held by the player.

- **Artwork:** Characters, scenes, and props are drawn programmatically, reducing the need for external image assets.

3 Experiments Run So Far and Their Results

The experiments conducted so far are engineering experiments. No experiments involving real students have been conducted yet.

Experiment	What Was Done	Result
Standalone Missions	Developed seven independent Easy missions with study packs.	The missions worked but felt like disconnected quizzes. This approach was replaced.
Connected Story	Rebuilt the application as a seven-chapter simulation with shared state.	All seven chapters are playable end-to-end, with money carrying across chapters.
Simulated Playthroughs	Ran two scripted players, one careful and one reckless, through the story logic.	Both players reach the ending. There are no dead ends or negative balances, and balances match diary entries.
Balance Check	Tested Rs. 1,500 with a deposit of Rs. 300, withdrawal of Rs. 200, and UPI payment of Rs. 150.	The final balance is Rs. 1,450, as specified.
Interest and Credit Calculations	Tested Rs. 2,000 over six months at 3% and 6%, and Rs. 3,000 store credit at 3% monthly interest with Rs. 300 minimum payments.	Interest is Rs. 30 and Rs. 60, respectively. Credit takes 12 payments and costs Rs. 3,497 in total, including Rs. 497 interest.
Affordability Check	Simulated a careful player saving Rs. 150 per week.	The player reaches approximately Rs. 890 by Chapter 5 and cannot afford the Rs. 1,200 laptop. This balance requires further review.
Reckless-path Outcome	Simulated a player who follows every scam and purchases on credit.	The player finishes with Rs. 0 in the bank and Rs. 3,258 owed, but can still complete the story.
Duplicate-tap Test	Tested two taps on the same button without an intervening frame.	A bug that charged a decision twice was identified and fixed in the controller. The test now passes.
Small-screen Telugu Layout	Tested the complete story at 360 × 640 resolution with 130% text scaling.	Initial layouts overflowed with the test font. After loading the actual font, all layouts fitted without overflow.

Experiment	What Was Done	Result
Visual Review	Rendered screenshots of every beat.	Approximately eight screens were visually inspected. The remaining screens still require review.

4 Progress on Testing

A total of 104 automated tests are passing.

4.1 Completed Tests

- **Financial rules:** Tested missing needs, payments made without sufficient funds, and complete story play-throughs. Save-format round-trip tests also pass.
- **Content validation:** Verified that all text keys exist in both languages with matching placeholders. No unfilled placeholders occur during the tested play-throughs, and no real URLs are present in the content.
- **Database:** Tested schema constraints, cascade deletes, upgrades from the old schema, rollback after failed migrations, profile isolation, resets, deletion of all data, and persistence after closing and reopening the application.
- **UI flow:** Tested onboarding in both languages, locked chapters, Chapter 1 activities, language switching while preserving progress, practice replay, and resuming after a simulated restart.
- **Full-story UI testing:** Played through all seven chapters four times using the real interface, covering careful and reckless choices, English and Telugu, and a small screen.
- **Static checks:** The code analyzer and formatter report no issues.

4.2 Pending Tests and Validation

- The debug APK has not yet been built on the current machine or installed on a physical Android device.
- The rewritten on-device test for Chapter 1, which uses real SQLite on Android, has not been run. No emulator is currently configured.
- Testing with real students has not been conducted. Therefore, the target of 5–10 minutes per chapter and the learning impact remain unmeasured.
- All Telugu story text remains an unreviewed draft and requires review by a fluent Telugu speaker.
- Performance on a budget Android phone has not been tested.

5 Plan for the Remaining Time

5.1 Medium and Advanced Levels

The existing game engine supports additional levels through new chapter scripts, rules, and text. The proposed division is as follows:

- **Medium:** Retain the seven topics while introducing competing goals and unexpected events, such as irregular income, shared family budgets, loan comparisons, and more convincing scams.
- **Advanced:** Introduce multi-step financial planning over longer periods, including compound interest, inflation, EMI affordability, insurance comparisons, and risk and diversification using fictional assets.
- **Implementation:** Each level requires scripts, rules with unit tests, any new activity screens, English and Telugu text, and full play-through testing. The onboarding screen already contains Medium and Advanced level options.
- **Open decision:** Determine whether these levels continue Asha's story or introduce separate stories.

5.2 Daily Check-in Notifications

Local scheduled notifications could support daily engagement without requiring a server or internet connectivity. This feature would introduce a plugin dependency, require notification permission on Android 13 and later, and need an on/off setting in the application. Since this changes the originally specified technology and permission scope, it should be documented in the report.

5.3 Daily Check-in Points

A database migration could introduce a per-profile record containing the last check-in date, streak count, and total points. Points could be displayed in the home-screen header and progress screen.

The check-in points should remain separate from chapter stars, which reward financial decision-making rather than attendance. The possibility of users manipulating streaks by changing the device clock should also be considered.

5.4 NGO Statistics Integration

Integrating statistics with an NGO is the largest proposed feature and conflicts with the application's current commitments to offline operation, no internet permission, and no personal data collection.

A full integration would require:

- An endpoint provided by the NGO.
- A student identifier recognised by the NGO.
- Appropriate consent wording and internet permission.

- An offline queue that synchronises data when connectivity becomes available.
- Secure handling of credentials and transmitted data.

Because the intended players are mostly minors, parental or school consent and data minimisation must be resolved with the NGO before implementation.

A lower-risk alternative is to export a progress summary as a file or QR code for collection by a teacher. This would provide statistics without requiring network connectivity or a server.

5.5 Prioritisation

The proposed implementation order is:

1. Implement daily check-in points and notifications.
2. Develop the Medium level.
3. Develop the Advanced level.
4. Implement NGO statistics export if full integration cannot be agreed upon in time.

Alongside these features, the remaining validation work includes building and installing the APK, running the on-device test, reviewing the Telugu translation, and testing performance on a budget Android device.